

CYBER-ZEB CONSULTING

BERANA LMS

End-to-End Product Concept, Functional Requirements,
Integration Architecture and Junior Developer Implementation Guide

For Higher Education, Schools, Businesses, Government, NGOs and Training Organizations

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Product Vision

Berana LMS will be a secure, modular and integration-ready learning platform that helps institutions manage learners, instructors, parents, courses, live virtual learning, payments, attendance, assessment, communication, reporting and institutional operations from one system.

Document Control

Field	Information
Product Name	Berana Learning Management System (Berana LMS)
Product Owner	Cyber-Zeb Consulting
Primary Audience	Internal software developers, technical leads, product owners, QA testers and implementation teams
Purpose	Provide a clear step-by-step blueprint for designing, developing, integrating, testing and launching Berana LMS
Delivery Approach	Modular MVP first, followed by controlled expansion
Classification	Confidential – Internal Use Only
Recommended Architecture	API-first modular monolith for MVP, designed for later service separation
Primary Platforms	Responsive web application first; mobile applications after MVP stabilization

How Junior Developers Should Use This Document

1. Read Sections 1–5 before writing code. They explain the product, users, scope and business workflows.
2. Use the user stories, acceptance criteria and data entities to create backlog items.
3. Develop one module at a time behind stable APIs; do not mix unrelated functions in one feature.
4. Use the MVP sequence in Section 18. Do not start advanced AI, IoT or analytics before the core workflows work.
5. Every completed feature must include validation, authorization, logging, tests, documentation and demonstration evidence.

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1. Berana LMS Concept and Business Purpose

Berana LMS is a multi-tenant digital learning platform designed to support formal education, corporate training, government workforce development, professional certification, NGO capacity-building and independent training providers. It combines self-paced learning, instructor-led teaching, live virtual classes, assessment, payments, reporting and communication within one controlled platform.

1.1 Problems Berana LMS Must Solve

- Learning information is often spread across spreadsheets, messaging applications, paper records and separate video-conferencing tools.
- Administrators lack one reliable view of enrollment, attendance, payments, performance and instructor activity.
- Learners need a consistent place to access courses, schedules, materials, assignments, grades and certificates.
- Instructors need practical tools to teach, assess, communicate and monitor progress without technical complexity.
- Parents need controlled visibility into a child's attendance, grades, behavior, announcements and teacher communication.
- Organizations need secure integration with identity, HR, student information, finance, payment, video, email and reporting systems.

1.2 Product Principles

Principle	Meaning for Development
Simple First	Core workflows must be easy for inexperienced users and low-bandwidth environments.
Role-Based	Every screen and API must enforce the user's role, organization and permissions.
Mobile Responsive	The web application must work well on phones, tablets and desktops.
API First	All major functions should be available through documented internal APIs.
Modular	Institutions can enable only the modules they need.
Multi-Tenant	Each institution's users, data, branding, settings and reports remain separated.
Secure by Design	Authentication, authorization, encryption, audit and privacy are built into every module.
Integration Ready	Use standard protocols and adapters for Zoom, payment, SSO, SIS, HR, email and learning tools.
Low-Bandwidth Aware	Support compressed media, resumable uploads, downloadable materials and asynchronous alternatives.

2. Target Markets and Deployment Models

Market	Primary Use Cases	Special Requirements
Universities and Colleges	Programs, semesters, courses, credits, faculty, students, grades, transcripts and research/training activities	Academic hierarchy, credit rules, registrar integration, large cohorts
Primary and Secondary Schools	Classes, subjects, teachers, students, parents, attendance, report cards and school communication	Parent portal, safeguarding, behavior and homeroom workflows
Businesses	Employee onboarding, compliance, technical training, leadership and role-based learning	HR integration, manager dashboards, mandatory training and competency tracking

Market	Primary Use Cases	Special Requirements
Government Institutions	Civil-service learning, policy awareness, professional development and certification	Strong audit, identity, reporting and data-residency controls
NGOs and International Organizations	Program training, beneficiary learning, partner capacity and field workforce development	Multi-language, offline/low-bandwidth, donor reporting
Training Providers	Public course catalog, enrollment, payment, instructor scheduling and certificates	E-commerce, coupons, taxes, refunds and marketing

2.1 Deployment Options

Option	Description	Recommended Use
Shared SaaS	Multiple institutions use one platform with isolated tenant data and branding.	Small and medium institutions; fastest rollout
Dedicated Cloud	A separate application/database environment for one customer.	Banks, government, universities or customers requiring isolation
Private Cloud / On-Premises	Installed in customer-controlled infrastructure.	Highly regulated or data-residency-sensitive customers
Hybrid	Core platform in cloud with selected identity, data or reporting systems on-premises.	Large institutions with legacy systems

3. User Roles and Permission Model

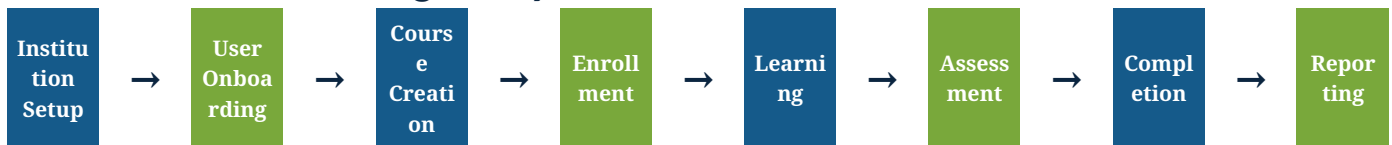
Role	Typical Owner	Key Permissions
Platform Super Administrator	Cyber-Zeb operations	Creates tenants, manages global settings, plans, integrations, support and platform health.
Institution Administrator	Institution/department	Manages the institution, users, branding, policies, modules, reports and integrations.
Academic Administrator / Registrar	Education institution	Manages programs, terms, courses, enrollment, grades, progression and academic records.
Training Administrator	Business/organization	Manages catalogs, cohorts, mandatory training, instructors and completion reporting.
Department / Campus Administrator	Delegated unit	Manages only assigned campus, school, faculty, department or business unit.
Instructor / Teacher	Teaching staff	Creates content, schedules classes, teaches, assesses, grades, communicates and monitors learners.
Teaching Assistant	Support staff	Supports classes and grading within delegated permissions.
Student / Learner	End user	Accesses courses, live sessions, materials, assignments, assessments, grades and certificates.
Parent / Guardian	Linked to child	Views permitted attendance, grades, reports, announcements and communicates with authorized teachers.
Manager / Supervisor	Business context	Views assigned employees' training, compliance, skills and development plans.
Finance Officer	Institution finance	Manages invoices, payments, refunds, discounts

Role	Typical Owner	Key Permissions
		and reconciliation.
Support / Help Desk	Operational support	Handles tickets and limited user troubleshooting without broad data access.
Auditor / Read-Only Reviewer	Compliance	Views selected records, logs and reports without changing data.

3.1 Permission Design Rules

- Use role-based access control plus tenant, campus, department, course and relationship scope.
- A parent can see only linked children and only information authorized by the institution.
- An instructor can access only assigned courses, classes and learners unless additional authority is granted.
- Finance data must be separated from academic permissions.
- Sensitive actions such as grade changes, refunds, user impersonation and integration-secret updates require audit logging and stronger approval.

4. End-to-End Learning Lifecycle



Stage	Main Activities	System Output
Institution Setup	Create tenant, branding, structure, policies, roles, integrations and academic calendar.	Operational institution workspace
User Onboarding	Invite/import users, verify identity, assign roles and link parents/managers.	Active user accounts
Course Creation	Create course, outcomes, modules, content, schedule, instructor and grading rules.	Published course
Enrollment	Self-enrollment, administrator enrollment, cohort assignment or payment-based enrollment.	Learner-course registration
Learning	Self-paced content, discussions, assignments and live Zoom sessions.	Learning activity records
Assessment	Quizzes, assignments, exams, rubrics and instructor grading.	Scores and feedback
Completion	Apply completion rules, issue certificate/credit and close course.	Completion record
Reporting	Attendance, performance, compliance, financial and operational reports.	Dashboards and exports

5. Functional Modules Overview

#	Module	Purpose
1	Tenant & Institution Management	Organizations, campuses, faculties, departments, programs, branding and configuration

#	Module	Purpose
2	Identity & Access	Registration, login, SSO, MFA, roles, permissions and account lifecycle
3	User Profiles	Students, instructors, parents, managers, staff and relationships
4	Academic / Training Structure	Academic year, term, program, level, section, cohort, curriculum and credit
5	Course Catalog & Authoring	Course templates, modules, lessons, outcomes, prerequisites and publishing
6	Content Management	Documents, video, audio, SCORM/xAPI packages, links and versioning
7	Enrollment & Cohorts	Manual, bulk, self, invitation, payment and rule-based enrollment
8	Virtual Classroom	Zoom scheduling, joining, attendance, recordings, webhooks and calendar
9	Attendance	Class/session attendance, manual correction, rules, alerts and reports
10	Assignments & Assessments	Assignments, quizzes, question bank, exams, rubrics, grading and feedback
11	Gradebook & Progress	Weighted grades, outcomes, progression, transcripts and completion
12	Communication	Announcements, messages, discussions, email/SMS/push and parent-teacher communication
13	Payments & Billing	Prices, invoices, checkout, discounts, refunds, reconciliation and access activation
14	Certificates & Credentials	Templates, verification, QR code, expiration and continuing education
15	Parent / Manager Portal	Linked learner visibility, reports, alerts, communication and acknowledgments
16	Reports & Analytics	Academic, training, engagement, compliance, instructor, finance and executive reporting
17	AI Services	Recommendations, assistance, risk alerts, content support and analytics
18	IoT / Physical Integration	Smart attendance, classroom devices, environmental data and campus events
19	Integration Hub & API	Zoom, payments, SSO, SIS/HR, email/SMS, LTI, OneRoster and webhooks
20	Administration & Support	Settings, audit, help desk, templates, localization, retention and backup

6. Identity, Organization and User Management

6.1 Institution Setup Workflow

- Platform administrator creates the institution tenant and unique tenant code.
- Institution administrator selects institution type, timezone, language, currency and academic/training model.
- Create campuses, faculties, schools, departments, branches or business units.
- Configure branding, domain, email sender, policies, grading, attendance and completion defaults.

10. Configure identity method: local account, Google/Microsoft SSO, SAML or OpenID Connect.
11. Create roles and delegated administrative scopes.
12. Import or invite users; validate duplicates and required attributes.
13. Activate integrations and run test transactions before production use.

6.2 User Account Fields

Entity	Minimum Fields	Optional / Advanced Fields
Person	ID, legal/display name, email or phone, status, locale, timezone	National/student/employee ID, preferred name, demographic fields based on policy
Student	Student number, institution, program/class/cohort, admission status	Guardian links, accommodations, scholarship, prior education
Instructor	Staff number, department, qualification, expertise, employment status	Biography, office hours, certifications, availability
Parent/Guardian	Person account, relationship type, linked learner, permission status	Preferred communication, legal custody restrictions
Employee Learner	Employee ID, business unit, manager, job role	Competency profile, mandatory learning plan

7. Academic Structure, Course and Content Management

7.1 Academic Hierarchy



7.2 Course Authoring Steps

14. Create course identity: code, title, description, category, level, language and owner.
15. Define learning outcomes and map each outcome to lessons and assessments.
16. Create modules or weeks in the required sequence.
17. Add lessons: text, file, video, audio, URL, live session, discussion, assignment or quiz.
18. Define prerequisites, release dates, due dates and completion rules.
19. Assign instructor, teaching assistants, capacity and enrollment method.
20. Configure grading, attendance and certificate/credit rules.
21. Preview as student, run accessibility and mobile checks, approve and publish.

7.3 Content Standards

Content Type	Required Capability	Developer Notes
Text / HTML	Rich text, images, equations, links and accessible headings	Sanitize HTML; store versions
Documents	PDF, Office files and downloadable resources	Virus scan; access control; file-size limits
Video / Audio	Streaming, captions, transcript and progress tracking	Use adaptive delivery where possible
SCORM / xAPI	Import packages and record completion/activity	Treat as a dedicated integration module
External Tool	Launch approved third-party learning tools	Prefer LTI 1.3 / LTI Advantage
Question Bank	Reusable questions, categories, difficulty and outcomes	Version questions and protect exam items

8. Enrollment, Student and Parent Portals

8.1 Enrollment Methods

Method	Workflow	Use Case
Administrator Enrollment	Admin selects users/cohort and course; validates capacity and prerequisites.	Formal programs and mandatory training
Self-Enrollment	Learner selects course and accepts rules; may require key or approval.	Open internal catalog
Payment Enrollment	Enrollment becomes active only after verified payment event.	Public paid courses
Invitation Enrollment	System sends time-limited invitation token.	Partner or selected groups
Rule-Based Enrollment	Automatically enroll based on program, job role, department or cohort.	Academic curricula and compliance training
Bulk Import / API	Import registrations through CSV or external SIS/HR API.	Large institutions

8.2 Student Portal

- Personal dashboard and upcoming activities
- Course catalog and enrolled courses
- Learning progress and deadlines
- Zoom class schedule and join button
- Assignments, quizzes, grades and feedback
- Attendance history
- Payments, invoices and receipts
- Certificates and verifiable credentials
- Messages, announcements and support tickets
- Profile, preferences and privacy controls

8.3 Parent / Guardian Portal

Feature	Description	Privacy Control
Child Overview	Current classes, attendance summary, grades and upcoming deadlines.	Only linked child and authorized fields
Attendance Alerts	Notify absence, late arrival or repeated attendance risk.	Institution-configurable thresholds
Report Cards	View published term reports and teacher comments.	Release-date and acknowledgment control
Teacher Communication	Secure message or appointment request to authorized teachers.	No exposure of private staff contact details
Announcements	School, class and emergency notices.	Targeted by child's institution/class
Payments	View permitted fees, invoices and receipts; make payment where enabled.	Financial visibility controlled by institution
Consent / Acknowledgment	Approve forms, policies or activities electronically.	Audit timestamp and evidence

9. Instructor, Teaching and Classroom Management

Instructor Capability	Required Functions
Dashboard	Today's classes, ungraded work, messages, attendance exceptions and at-risk learners
Course Authoring	Create or reuse content within institutional approval rules
Class Management	Roster, groups, seating/sections, schedules and announcements
Live Teaching	Create/join Zoom sessions, share materials and view attendance data
Assessment	Question bank, quiz, assignment, rubric, grading, feedback and regrade workflow
Progress Monitoring	Learner progress, outcome mastery, inactivity and intervention notes
Communication	Announcements, discussions, individual messages and parent communication
Office Hours	Availability slots, appointment requests and meeting links
Reports	Class attendance, grade distribution, completion and engagement

10. Virtual Learning and Zoom Integration

Recommended Integration Choice

Use Zoom OAuth and REST APIs to create and manage meetings, Zoom webhooks to receive meeting events, and optionally the Zoom Meeting SDK to embed the meeting experience inside Berana. Start the MVP with secure meeting creation and join links; embed the Meeting SDK only after the basic workflow is stable.

10.1 Zoom Integration Workflow

22. Create and configure the Berana Zoom application in the Zoom developer platform.
23. Use OAuth so an authorized institution or instructor can connect a Zoom account.
24. Store access and refresh tokens encrypted; rotate and revoke them safely.
25. When an instructor schedules a live session, Berana calls the Zoom Meetings API.
26. Store the Zoom meeting ID, host account, start time, timezone, join URL and session mapping.
27. Show role-appropriate join/start controls in Berana; never expose host secrets to students.
28. Receive and verify Zoom webhooks for meeting start, end, participant join/leave and recording events.
29. Convert verified participant events into attendance evidence using configurable rules.
30. Retrieve and link cloud recordings only when permitted by institution policy and consent.
31. Handle retries, duplicate webhooks, token expiration, meeting cancellation and changed schedules.

10.2 Zoom Data Mapping

Berana Entity	Zoom Object / Event	Purpose
Live Session	Meeting	Schedule and identify virtual class
Instructor Host	Zoom user / host authorization	Create, start and manage meeting
Learner Participant	Participant event identity	Attendance and engagement evidence
Session Attendance	Join/leave webhook events	Calculate attended minutes and status
Recording Resource	Cloud recording event/file	Publish approved replay to course
Calendar Entry	Meeting start/end/timezone	Show schedule and reminders

10.3 Zoom Security Rules

- Use waiting room, passcode and authenticated-user settings based on institution policy.
- Do not place OAuth client secrets, SDK secrets or access tokens in browser code.
- Verify webhook signatures and make processing idempotent.
- Restrict recordings, transcripts and chat access by course role and retention policy.
- Maintain an audit record for meeting creation, update, cancellation and recording publication.

11. Attendance, Assessment, Gradebook and Certification

11.1 Attendance Sources

Source	How It Works	Control
Instructor Manual	Instructor marks present, absent, late, excused or left early.	Correction history and approval
Zoom Events	System calculates presence from join/leave data.	Identity matching and configurable duration rule
QR Code	Time-limited class QR scanned by learner.	Device/session validation and fraud controls
IoT / Smart Device	Approved reader or classroom device sends attendance event.	Signed device identity and secure API
Bulk Import	Authorized administrator uploads attendance file.	Validation and audit

11.2 Assessment Types

Type	Core Features
Quiz	Multiple-choice, true/false, matching, short answer, randomization, timer and attempts
Assignment	File or text submission, due date, late policy, rubric, feedback and resubmission
Exam	Secure window, question pools, timing, attempt control and proctoring integration option
Practical / Observation	Checklist or rubric completed by evaluator
Discussion / Participation	Participation grading based on configured criteria
External Result	Authorized import from another system or certification body

11.3 Gradebook Rules

- Support points, percentage, letter grade, pass/fail and competency/outcome-based grading.
- Support categories and weighted calculation.
- Allow draft, moderated and published grade states.
- Record every grade change with old value, new value, user, time and reason.
- Provide incomplete, excused and not-applicable handling.
- Support term report, transcript, completion and certificate rules.

12. Communication, Notifications and Support

Channel	Examples	Integration
In-App	Announcements, messages, alerts, tasks and notifications	Internal notification service
Email	Enrollment, reminders, payment receipt, grades and announcements	SMTP or email provider API
SMS	Urgent alerts, absence, OTP and deadline reminders	Provider adapter by country
Push Notification	Mobile reminders and updates	Firebase/APNs after mobile phase
Discussion	Course forums and moderated Q&A	Internal module
Parent-Teacher Messaging	Secure relationship-scoped communication	Internal module with audit
Help Desk	Support ticket, category, priority, assignment and resolution	Internal or external ticketing API

12.1 Notification Design

32. Create events such as enrollment.completed, class.reminder, attendance.absent and payment.succeeded.
33. Send events to a notification service that applies templates, language, channel and user preference.
34. Queue delivery so a slow provider does not block the main transaction.
35. Record delivery status, retries, failure reason and user read/acknowledgment where applicable.

13. Payments, Billing and Financial Controls

Payment Architecture

Build one internal Payment Service and multiple provider adapters. The rest of Berana should call the internal service rather than directly calling Stripe, PayPal, Chapa, a bank or a mobile-money provider. This allows providers to be changed by country or customer.

13.1 Payment Use Cases

- One-time course purchase
- Program or semester fee
- Subscription plan
- Installment payment
- Institution-sponsored enrollment
- Coupon, scholarship or discount
- Invoice and manual bank-payment confirmation
- Refund, cancellation and credit

13.2 Secure Payment Workflow



Step	Developer Action	Important Rule
Create Order	Create internal order with amount, currency,	Never trust amount sent from browser

Step	Developer Action	Important Rule
	payer, items and expiration.	
Create Checkout	Server requests hosted/embedded provider checkout.	Use provider tokenization; do not store card details
Receive Result	User returns to Berana after checkout.	Return page alone does not prove payment
Verify Webhook	Validate provider signature and event.	Idempotent processing required
Update Payment	Mark paid, failed, refunded or disputed.	Maintain immutable transaction history
Grant Access	Activate enrollment only after trusted success event.	Prevent duplicate enrollment
Reconcile	Match provider settlement and internal records.	Finance dashboard and exception report

13.3 Financial Data Entities

- Product/price
- Order
- Invoice
- Payment attempt
- Payment transaction
- Refund
- Discount/coupon
- Scholarship/sponsor
- Tax/fee
- Receipt
- Settlement/reconciliation record

14. Reporting, Analytics, AI and IoT Capabilities

14.1 Required Reports

Audience	Reports
Executive / Institution Admin	Enrollment, active users, completion, revenue, attendance, learner risk, instructor activity and platform usage
Academic / Training Admin	Course capacity, progression, grade distribution, dropout, completion and certification
Instructor	Roster, attendance, engagement, assignment status, grades and at-risk learners
Student	Progress, attendance, grades, deadlines, competencies and certificates
Parent / Guardian	Authorized child attendance, grades, reports, notices and teacher communication
Manager / HR	Employee compliance, assigned learning, completion, overdue training and skills
Finance	Orders, payments, refunds, outstanding invoices, discounts and reconciliation
Support / Operations	Incidents, tickets, integration failures, job queues and availability

14.2 AI Features – Phased and Controlled

Phase	AI Capability	Guardrails
Later MVP / Phase 2	Search assistant over approved course content	Cite source content; role-aware access; no hidden student data
Phase 2	Learner risk indicators based on inactivity, attendance and assessment	Explain factors; human review; no automatic punishment
Phase 2	Personalized content or course recommendations	Allow opt-out; avoid sensitive profiling
Phase 3	Instructor assistance for quiz drafts, summaries and feedback suggestions	Instructor approval required; plagiarism and quality checks
Phase 3	Natural-language reporting and analytics	Apply tenant and role filters before AI processing
Phase 3	Translation, caption and transcript assistance	Human review for high-stakes academic content

14.3 IoT and Physical Integration Options

Integration	Example	API / Security Requirement
Smart Attendance Device	RFID/NFC/biometric reader sends attendance event	Device registration, signed requests, timestamp, anti-replay and privacy approval
Classroom Display / Kiosk	Shows timetable, announcements or QR attendance	Read-only scoped token and remote management
Environmental Sensor	Temperature, occupancy or air-quality data for learning spaces	IoT gateway; event ingestion; retention limits
Access Control	Course/class eligibility may be used as a reference for room access	Do not make LMS the only life-safety control; explicit integration boundary
Library / Lab Equipment	Training or course eligibility controls reservation	API-based entitlement and audit

15. Integration and API Architecture

Core Integration Principle

Berana owns its internal business data. External systems are connected through versioned APIs, webhooks, adapters and scheduled synchronization. Every integration must define system of record, data ownership, authentication, failure handling and reconciliation.

15.1 Integration Categories

Category	Recommended Interface	Purpose
Identity / SSO	OpenID Connect, OAuth 2.0, SAML, Microsoft Entra ID, Google Workspace	Authentication and account lifecycle
Virtual Classroom	Zoom REST API, OAuth, webhooks and optional Meeting SDK	Live classes, attendance and recordings
Payment	Stripe, PayPal, Chapa, mobile money or bank adapters	Checkout, payment confirmation, refunds and reconciliation

Category	Recommended Interface	Purpose
Student Information / ERP	OneRoster, CSV, REST APIs or institution-specific adapters	Users, courses, classes, enrollment and grades
HR / Workforce	REST API, SCIM or scheduled import	Employees, managers, job roles and mandatory training
Learning Tools	LTI 1.3 / LTI Advantage	Secure launch of third-party content/tools and grade return
Content Packages	SCORM and xAPI	Portable content and activity tracking
Communication	SMTP/email API, SMS provider and push service	Notifications and OTP
Calendar	Google Calendar, Microsoft Graph or iCalendar feeds	Schedules and reminders
Analytics / BI	Read replica, warehouse export or BI connector	Institutional dashboards
File Storage	S3-compatible object storage and CDN	Course files, submissions and recordings

15.2 Internal API Design

Area	Example Endpoints
Authentication	POST /api/v1/auth/login; POST /auth/refresh; POST /auth/logout
Users	GET/POST /api/v1/users; PATCH /users/{id}; POST /users/import
Organizations	GET/POST /tenants; /campuses; /departments; /programs
Courses	GET/POST /courses; /courses/{id}/modules; /lessons; /publish
Enrollments	POST /enrollments; GET /learners/{id}/enrollments; POST /enrollments/bulk
Live Sessions	POST /courses/{id}/live-sessions; POST /live-sessions/{id}/join
Attendance	POST /sessions/{id}/attendance; GET /attendance/reports
Assessments	POST /quizzes; /assignments; /submissions; /grades
Payments	POST /orders; POST /payments/checkout; POST /webhooks/payments/{provider}
Reports	GET /reports/completion; /attendance; /finance; /engagement
Integrations	POST /integrations/{type}/connect; /sync; /test; /disconnect

15.3 API Standards for Junior Developers

- Use HTTPS and JSON; version public APIs under /api/v1.
- Use stable resource names, correct HTTP methods and consistent error objects.
- Validate every request on the server; never trust tenant_id, role, amount or ownership from the client.
- Use pagination, filtering and sorting for list endpoints.
- Use idempotency keys for payment, enrollment and webhook-sensitive operations.
- Return correlation IDs and log them for support.
- Document APIs with OpenAPI and maintain examples.
- Apply rate limiting, authentication and authorization at the API gateway/application layer.

15.4 Webhook Processing Pattern



- Store provider event ID and reject duplicates.
- Do not perform slow work before returning the provider response.
- Retry failed processing with controlled backoff.
- Create a dead-letter queue and operations dashboard for repeated failures.

15.5 System-of-Record Matrix

Data	Recommended System of Record	Berana Responsibility
Institution LMS settings	Berana	Create and govern
Learner academic identity	SIS or Berana, by deployment	Synchronize with clear ownership
Employee identity/job role	HR system where available	Consume and map
Course learning content	Berana or approved content repository	Manage access and versions
Zoom meeting media	Zoom, linked from Berana	Store metadata and permissions
Payment transaction truth	Payment provider plus Berana ledger	Verify and reconcile
Grades	Berana or SIS, defined per institution	Prevent conflicting writes

16. Security, Privacy, Compliance and Audit

Control Area	Minimum Requirement
Authentication	Strong password rules, MFA for administrators, secure reset and session controls
Authorization	Tenant- and relationship-aware permissions on every API and query
Encryption	TLS in transit; encrypted databases, backups, secrets and tokens at rest
Secrets	Central secret manager; no secrets in source code or browser
Audit Logging	Login, role, grade, payment, user, integration and sensitive-data changes
Privacy	Data minimization, consent/notice, retention, access and deletion workflows
Child / Parent Data	Guardian relationship verification and institution-approved visibility
File Security	Virus/malware scan, type limits, private storage and signed URLs
Secure Development	Code review, dependency scanning, SAST/DAST and vulnerability remediation
Availability	Backups, restore testing, monitoring, queues, retries and disaster recovery
Integration Security	OAuth/OIDC, signed webhooks, scoped tokens, rotation and revocation
Data Isolation	Tenant isolation testing and prevention of cross-tenant queries

16.1 High-Risk Actions Requiring Audit

- Grade change after publication
- Manual attendance correction
- Refund or payment status override
- Parent-child relationship creation/change

- User impersonation
- Role or permission change
- Course completion/certificate override
- Integration connection or secret change
- Bulk export of learner data

17. Recommended Technical Architecture and Data Model

17.1 MVP Architecture

Recommended Starting Point

Build a modular monolith, not many microservices. Use one deployable backend with clear modules, one relational database with tenant controls, one job queue, object storage and external integration adapters. Separate services only after scale or operational need is proven.



Layer	Recommended Responsibility
Frontend	Responsive web portal for administrator, instructor, learner, parent and finance roles
Backend API	Authentication, authorization, business workflows and integration orchestration
Relational Database	Tenants, users, courses, enrollment, attendance, grades, payments and audit
Object Storage	Course files, submissions, certificates and approved media metadata
Cache	Sessions, frequently used settings and short-lived data
Queue / Workers	Email/SMS, reports, webhook processing, file scanning and sync jobs
Integration Adapters	Zoom, payments, SSO, email/SMS, SIS/HR and external tools
Observability	Logs, metrics, traces, alerts and audit search

17.2 Core Data Entities

Domain	Key Entities
Organization	Tenant, Campus, Department, Program, AcademicTerm, Cohort
Identity	User, Role, Permission, UserRole, GuardianLink
Learning	Course, CourseOffering, Module, Lesson, ContentItem, Enrollment
Delivery	LiveSession, AttendanceEvent, AttendanceRecord, LearningProgress
Assessment	Assignment, Quiz, Question, Submission, GradeItem, Grade
Credential	Certificate, verification code and expiration
Commerce	Order, Invoice, Payment, Refund, Discount
Communication	Message, Announcement, Notification, Discussion
Integration / Control	IntegrationConnection, WebhookEvent, AuditLog, SupportTicket

Domain	Key Entities

17.3 Multi-Tenant Rule

- Every tenant-owned table must carry tenant_id or use an equivalent strong isolation design.
- Every repository/query must apply tenant scope from the authenticated server context, not from user input.
- Automated tests must attempt cross-tenant access and prove it is denied.
- Global platform administrators require separately audited privileged access.

18. MVP Development Roadmap

Phase	Scope	Exit Criteria
Phase 0 – Discovery & Design	User roles, workflows, requirements, wireframes, architecture, data model, security and backlog	Approved product requirements and clickable key-screen prototype
Phase 1 – Platform Foundation	Tenant, users, roles, login, profile, organization structure, audit and basic administration	Administrators can securely create and manage an institution
Phase 2 – Core Learning	Course catalog, authoring, content, enrollment, learner/instructor portals and progress	A learner can enroll and complete a basic course
Phase 3 – Teaching & Assessment	Assignments, quizzes, submissions, gradebook, attendance and reports	Instructor can teach, assess and publish results
Phase 4 – Zoom & Communication	Zoom scheduling/joining/webhooks, announcements, email and reminders	Live class workflow and attendance work end to end
Phase 5 – Payment & Certificates	Orders, checkout adapter, verified webhooks, enrollment activation, receipts and certificates	Paid enrollment and verifiable completion work securely
Phase 6 – Parent / Manager Portal	Guardian links, authorized reports, alerts and teacher communication	Parent can securely view only linked learner information
Phase 7 – Hardening & Pilot	Performance, security, backup, accessibility, user acceptance, support and pilot migration	Approved pilot launch with operational runbook

18.1 MVP Must Include

- Multi-tenant institution setup and branding
- Role-based administrator, instructor, learner and parent accounts
- Course creation, content, enrollment and progress
- Assignments, quizzes, grades and attendance
- Zoom meeting creation, joining and event-based attendance
- Announcements, messaging and email notifications
- Payment provider adapter with verified webhook and enrollment activation
- Parent view of authorized attendance, grades and reports
- Basic dashboards, exports, audit logs and support tools

18.2 Features to Delay Until Core Stability

- Advanced AI tutoring
- Complex microservices
- Blockchain credentials
- Full offline mobile synchronization
- Advanced proctoring
- Predictive analytics with sensitive profiling
- Large marketplace/ecosystem

- Multiple payment providers at the same time

19. Sprint-by-Sprint Junior Developer Instructions

Sprint	Focus	Developer Tasks	Demonstration
Sprint 0	Requirements and setup	Read workflows; prepare stories; create repository, branching, environments, CI, coding standards, OpenAPI and database migration process.	Approved backlog and running skeleton
Sprint 1	Tenant and identity	Tenant, organization, user, role, permission, login, password reset and audit.	Secure admin login and tenant setup
Sprint 2	Course foundation	Course, offering, module, lesson, content metadata, publish states and catalog.	Admin/instructor publishes a course
Sprint 3	Enrollment and portals	Enrollment, dashboard, roster, progress and parent link foundation.	Learner sees enrolled course
Sprint 4	Assessment	Assignments, submissions, quiz basics, grade items and feedback.	Instructor grades learner work
Sprint 5	Attendance and schedules	Class sessions, manual attendance, calendar and reports.	Attendance captured and reported
Sprint 6	Zoom integration	OAuth connection, meeting creation, join flow, webhook receiver and attendance events.	Live session works in test account
Sprint 7	Communication	Announcements, messages, templates, email queue and preferences.	Role-based communication works
Sprint 8	Payments	Orders, price, checkout adapter, webhook verification, ledger and enrollment activation.	Test payment grants access once
Sprint 9	Parent portal and reports	Guardian scope, attendance/grade report, teacher messaging and acknowledgment.	Parent sees only linked child
Sprint 10	Certificates and hardening	Completion, certificate, verification, security tests, backup, monitoring and pilot fixes.	Pilot release candidate

19.1 Definition of Done for Every Story

- Acceptance criteria passed.
- Authorization tested for allowed and denied roles.
- Tenant isolation tested.
- Input validation and error handling implemented.
- Audit/logging added where required.
- Unit and integration tests added.
- API documentation updated.
- Database migration and rollback considered.
- Responsive UI and accessibility checked.
- Demo evidence and code review completed.

20. Testing, Deployment, Support and Operations

Test Type	Examples
Unit	Grade calculation, completion rule, payment state transition and permission function
API Integration	Create course, enroll learner, submit assignment and process webhook
End-to-End	Paid enrollment to course access; Zoom session to attendance; grade publication to parent view
Security	Cross-tenant access, privilege escalation, injection, file upload and broken object authorization
Performance	Large roster, report export, concurrent quiz and webhook burst
Usability	Administrator, instructor, learner and parent task completion
Accessibility	Keyboard, labels, contrast, headings, captions and screen-reader basics
Recovery	Database restore, object restore, queue replay and integration reconnection

20.1 Environments

Environment	Purpose	Data Rule
Development	Individual/team coding and automated tests	Synthetic data only
Test / QA	Integration and regression testing	Synthetic or anonymized data
Staging / UAT	Customer-like validation and training	Controlled test data
Production	Live institution use	Restricted, monitored and backed up

20.2 Operations Dashboard

- Application availability and error rate
- API latency and failed requests
- Queue depth and failed jobs
- Zoom token/webhook status
- Payment webhook and reconciliation exceptions
- Email/SMS failures
- Storage and database health
- Security alerts and privileged actions
- Backup completion and restore-test status

21. KPIs and Product Success Measures

Area	KPI	Example Direction
Adoption	Monthly active learners and instructors	Growing by institution
Learning	Course completion rate	Measured by course and cohort
Engagement	Active learning days, content completion and live-session participation	Identify low engagement
Attendance	Attendance rate and unverified events	High reliability; low exception count

Area	KPI	Example Direction
Assessment	Submission and grading turnaround	Defined service level
Parent Engagement	Parent logins, report views and teacher interactions	Institution-defined
Commerce	Checkout conversion, payment success and reconciliation exceptions	High success; zero unexplained mismatch
Operations	Availability, response time and support resolution	SLA targets
Security	Critical vulnerabilities, unauthorized access and audit exceptions	Zero open critical issues
Integration	Webhook success, sync failures and retry backlog	Monitored and corrected

22. Risks, Assumptions and Future Enhancements

Risk	Impact	Mitigation
Scope too large	Delayed or failed MVP	Strict phase gates and product-owner approval
Junior team complexity	Quality and security weaknesses	Architecture review, coding standards, mentoring and small stories
Zoom/payment API changes	Integration failure	Adapters, version monitoring, sandbox tests and operational alerts
Poor data ownership	Duplicate/conflicting records	System-of-record matrix and reconciliation
Weak authorization	Privacy breach	Central policy, automated tests and security review
Low bandwidth	Poor learner experience	Compressed content, asynchronous options and performance budget
Parent relationship error	Unauthorized child-data access	Verified linking, approval, audit and revocation
Payment mismatch	Access or financial loss	Verified webhooks, idempotency and reconciliation
AI misuse	Bias, incorrect decisions or privacy risk	Human approval, explainability and phased introduction

22.1 Future Enhancements

- Native Android/iOS apps
- Offline learning packs and synchronization
- Advanced competency and talent marketplace
- Adaptive learning paths
- Digital credential wallets
- Advanced proctoring integrations
- Library, hostel, transport and campus-system integration
- Marketplace for courses and instructors
- Multilingual AI assistant
- Data warehouse and institution benchmarking

Appendix A – Sample User Stories

Role	User Story	Acceptance Criteria
Student	As a learner, I want to see all upcoming classes and deadlines so that I can plan my study.	Only enrolled-course items appear; timezone applied; completed items identifiable.
Instructor	As an instructor, I want to schedule a Zoom class from my course so learners can join from Berana.	Meeting created through authorized account; course roster can join; audit recorded.
Parent	As a parent, I want to see my child's attendance so I can follow up on absences.	Only linked child appears; published records only; access logged.
Finance	As a finance officer, I want to reconcile successful payments with enrollments.	Provider transaction, internal order, amount and enrollment status match.
Administrator	As an administrator, I want to import users and courses from another system.	Validation report, duplicate handling and audit provided.

Appendix B – Initial API Event Catalog

Event	Typical Consumers
tenant.created	Notification service, analytics, audit, integration worker or reporting
user.invited	Notification service, analytics, audit, integration worker or reporting
user.activated	Notification service, analytics, audit, integration worker or reporting
guardian.linked	Notification service, analytics, audit, integration worker or reporting
course.published	Notification service, analytics, audit, integration worker or reporting
enrollment.created	Notification service, analytics, audit, integration worker or reporting
enrollment.completed	Notification service, analytics, audit, integration worker or reporting
live_session.created	Notification service, analytics, audit, integration worker or reporting
live_session.started	Notification service, analytics, audit, integration worker or reporting
participant.joined	Notification service, analytics, audit, integration worker or reporting
participant.left	Notification service, analytics, audit, integration worker or reporting
attendance.finalized	Notification service, analytics, audit, integration worker or reporting
assignment.submitted	Notification service, analytics, audit, integration worker or reporting
grade.published	Notification service, analytics, audit, integration worker or reporting
payment.succeeded	Notification service, analytics, audit, integration worker or reporting
payment.failed	Notification service, analytics, audit, integration worker or reporting
payment.refunded	Notification service, analytics, audit, integration worker or reporting
certificate.issued	Notification service, analytics, audit, integration worker or reporting
notification.failed	Notification service, analytics, audit, integration worker or reporting

Appendix C – Integration Readiness Checklist

- Business owner and technical owner named

- Purpose, data and system of record defined
- API documentation and sandbox available
- Authentication and scopes defined
- Secrets storage and rotation defined
- Webhook verification and idempotency designed
- Rate limits, retries and timeouts defined
- Field mapping and timezone/currency rules documented
- Failure, reconciliation and manual recovery process documented
- Security, privacy and retention approved
- Test cases and monitoring alerts prepared

Appendix D – Official Technical References Used

Source	How It Informed This Blueprint
Zoom Developer Documentation	Meeting SDK, OAuth integrations, meeting webhooks and authorization requirements
Stripe Documentation	Hosted/embedded Checkout, Checkout Sessions and webhook-based fulfillment
1EdTech Standards	LTI 1.3/LTI Advantage for learning-tool integration and OneRoster 1.2 for roster/grade exchange
Moodle Documentation	Reference model for parent/mentor role visibility

Appendix E – First Team Instruction

Start Here

Do not begin by building every screen. First complete the domain model, role/permission matrix, key workflows, API contract and clickable prototype. Then build the MVP in the phase order shown. Demonstrate one complete end-to-end flow at the end of every sprint.

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